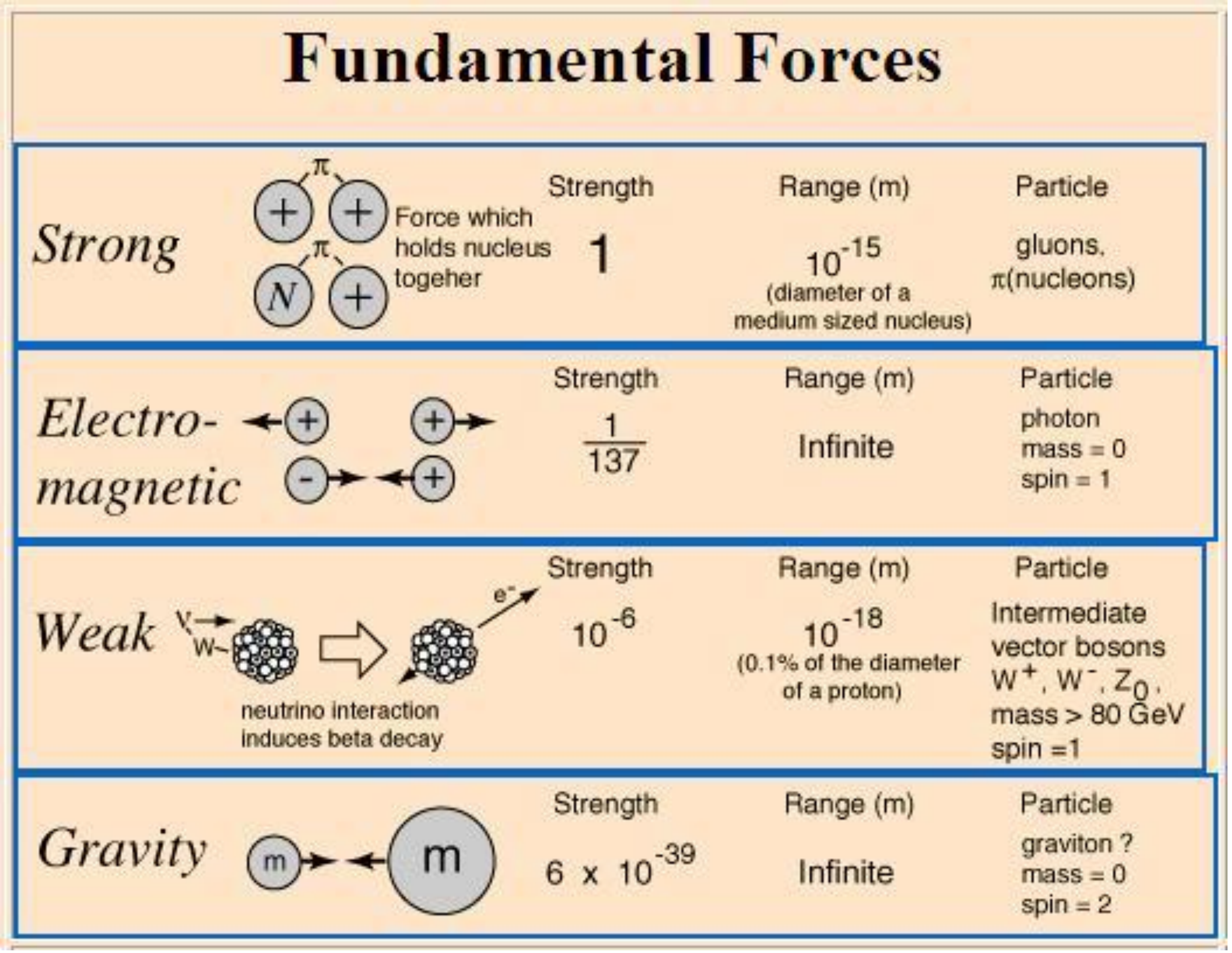
Introduction

Electromagnetic Fields and Waves

Electromagnetic interaction is ubiquitous



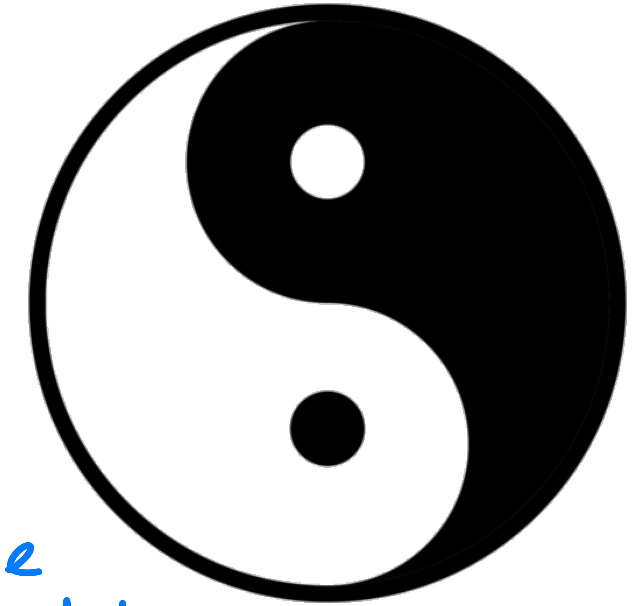
Fundamental Forces

		Strength	Range (m)	Particle
Strong	Force which holds nucleus togeher	1	10^{-15} (diameter of a medium sized nucleus)	gluons, π(nucleons)
Electro-magnetic		$\frac{1}{137}$	Infinite	photon mass = 0 spin = 1
Weak	neutrino interaction induces beta decay	10^{-6}	10^{-18} (0.1% of the diameter of a proton)	Intermediate vector bosons W^+, W^-, Z_0, mass > 80 GeV spin =1
Gravity		6×10^{-39}	Infinite	graviton ? mass = 0 spin = 2

we can smell!

Electricity and magnetism

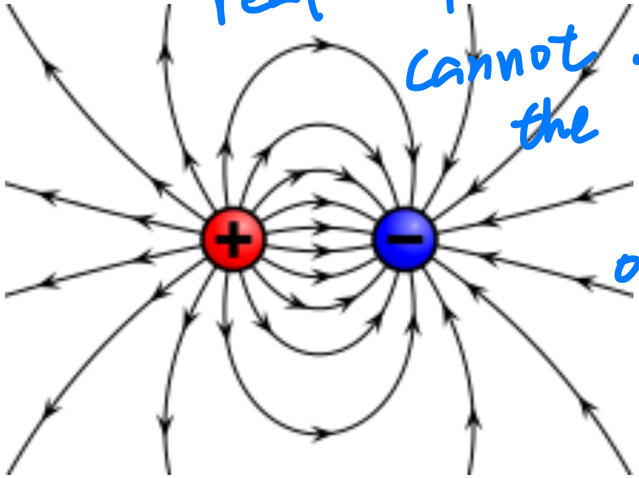
Electricity



Magnetism

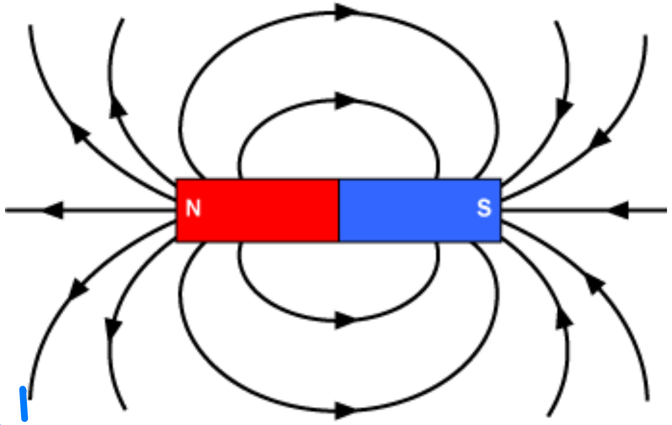
real field

dipole!



cannot see the field lines!

or place metal dusts, will align field lines



similar!

Electric field: $\vec{E}$

Light is EM field

Magnetic field: $\vec{B}$

Maxwell's Equations

Will learn in the entire course!

Name	Integral equations	Differential equations
Gauss's law	$\oiint_{\partial\Omega} \mathbf{E} \cdot \mathrm{d}\mathbf{S} = \frac{1}{\varepsilon_0} \iiint_{\Omega} \rho \, \mathrm{d}V$	divergence $\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$ ← electron density at this particular this time & space!
Gauss's law for magnetism	$\oiint_{\partial\Omega} \mathbf{B} \cdot \mathrm{d}\mathbf{S} = 0$	$\nabla \cdot \mathbf{B} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\oint_{\partial\Sigma} \mathbf{E} \cdot \mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \iint_{\Sigma} \mathbf{B} \cdot \mathrm{d}\mathbf{S}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampère's circuital law (with Maxwell's addition)	$\oint_{\partial\Sigma} \mathbf{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 \left(\iint_{\Sigma} \mathbf{J} \cdot \mathrm{d}\mathbf{S} + \varepsilon_0 \frac{\mathrm{d}}{\mathrm{d}t} \iint_{\Sigma} \mathbf{E} \cdot \mathrm{d}\mathbf{S} \right)$	$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$

→ difficult to solve!

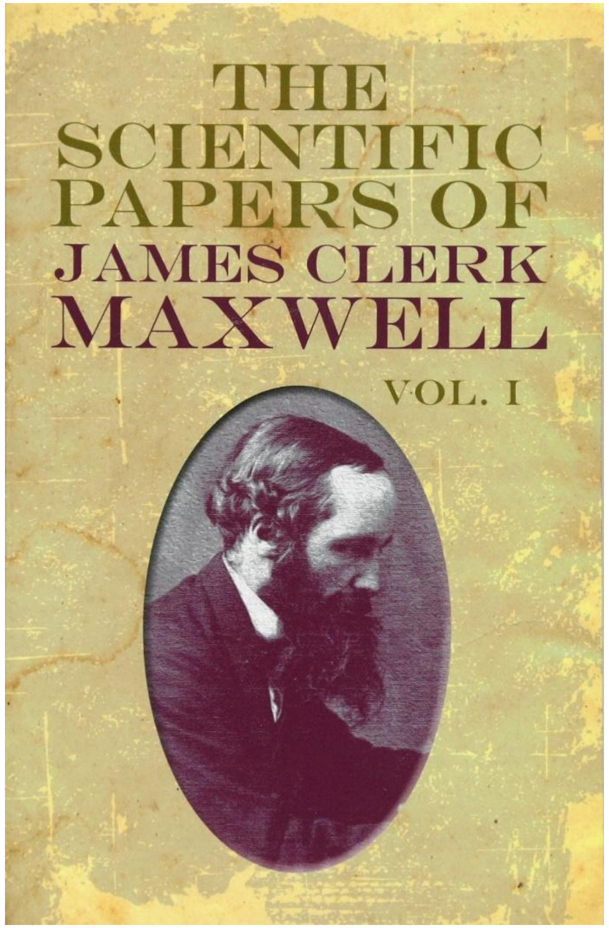
- A **set** of **coupled** linear **partial** differential equations of **vectors!**
- They dictate everything about classical electromagnetism.

↓ 8 equation

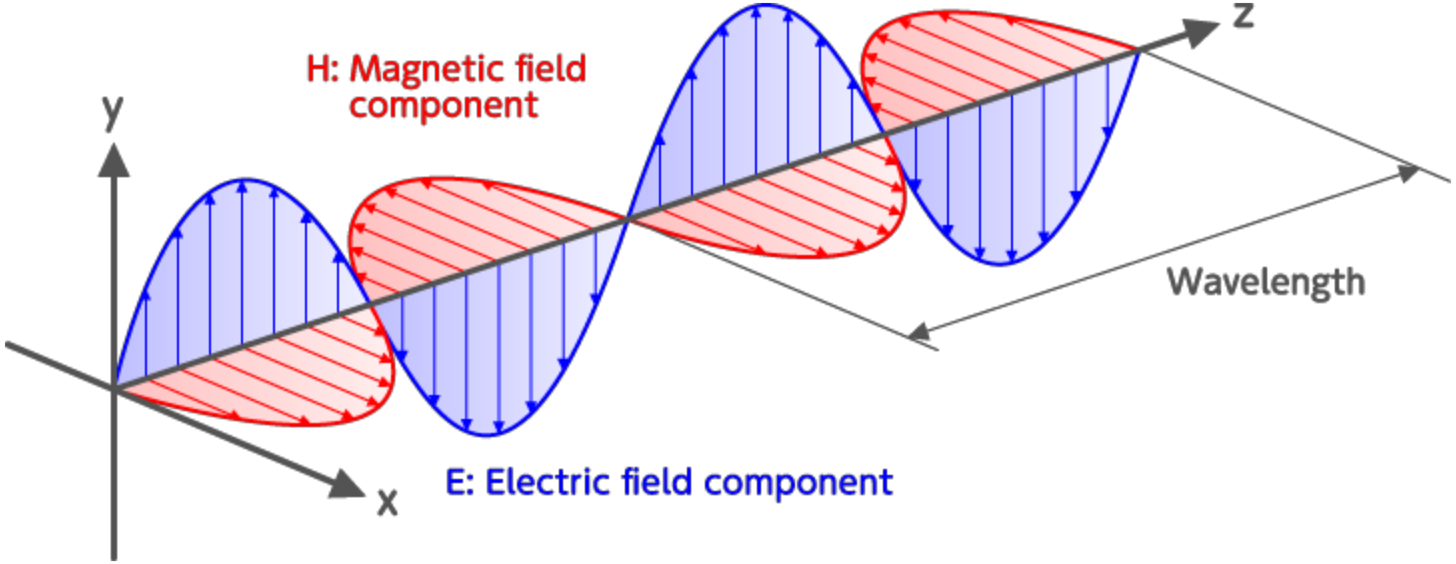
actually total 20!

A glorious moment in science history

will spend one entire lecture for that!



Published 1856 ~ 1862



light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena *--James Clerk Maxwell*

It is a “classical” theory

- “Classical” means: it does not incorporate quantum principles (e.g., photon quantization)
- “Classical” also means: it lays the foundation for modern physics. **Maxwell theory introduced:**
    - Field as fundamental entity
    - Local differential equations
    - Finite propagation speed
    - Lorentz invariance
    - Gauge redundancy and gauge invariance

in deep:

empty with

charge / magnet

↓

field

They are **structural pillars of physics**.

↳ important concepts

It is a “classical” theory (contd.)

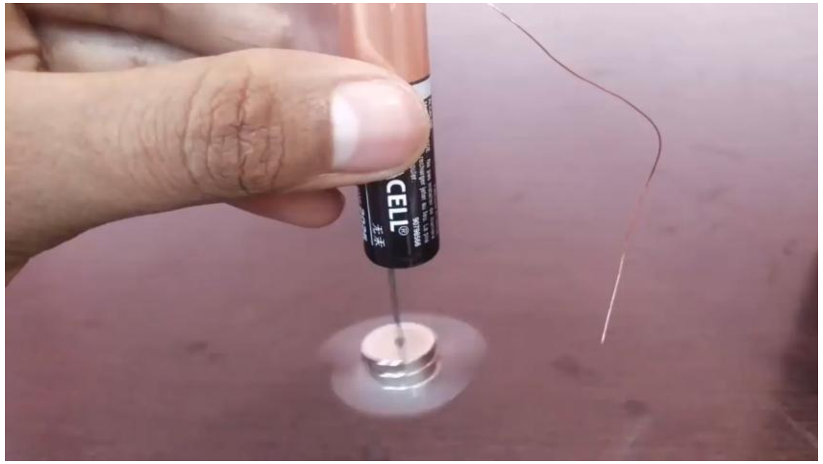
- **Maxwell theory directly led to:**
  - Special Relativity
  - The concept of relativistic field theory
- **Maxwell theory later generalized into:**
  - Quantum Electrodynamics
  - Gauge theory
- https://www.bilibili.com/video/av17923311/
- Chen Ning Yang, “The conceptual origins of Maxwell’s equations and gauge theory”, Physics Today **67**, 11, 45 (2014)
  - Gauge symmetry originates in electromagnetism, not quantum mechanics.
  - Maxwell theory introduced local gauge invariance — the organizing principle of modern particle physics.

Join the tour of Maxwell's wonder world

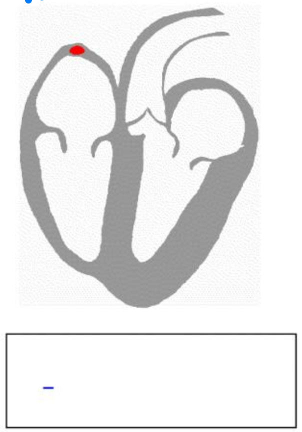
Maxwell was a religious person. I wonder whether he had in his prayers asked for God's forgiveness for revealing one of His greatest secrets.

-- Chen Ning Yang

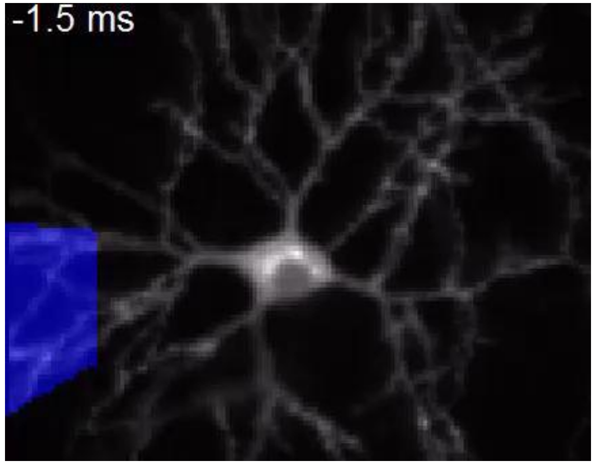
Interesting EM phenomena

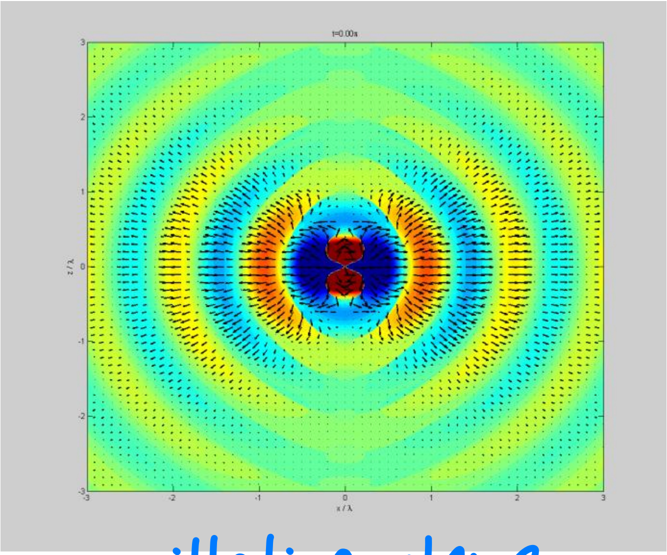


heartbeat

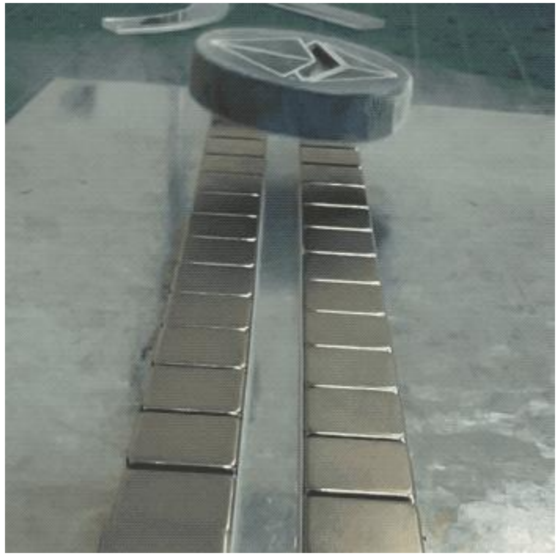


neurons



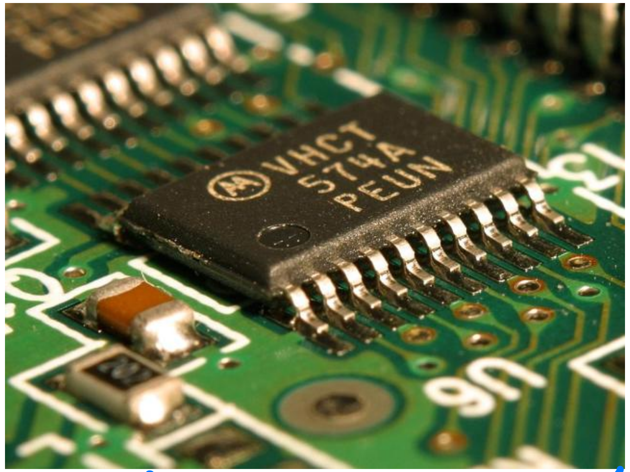


oscillating charge



float

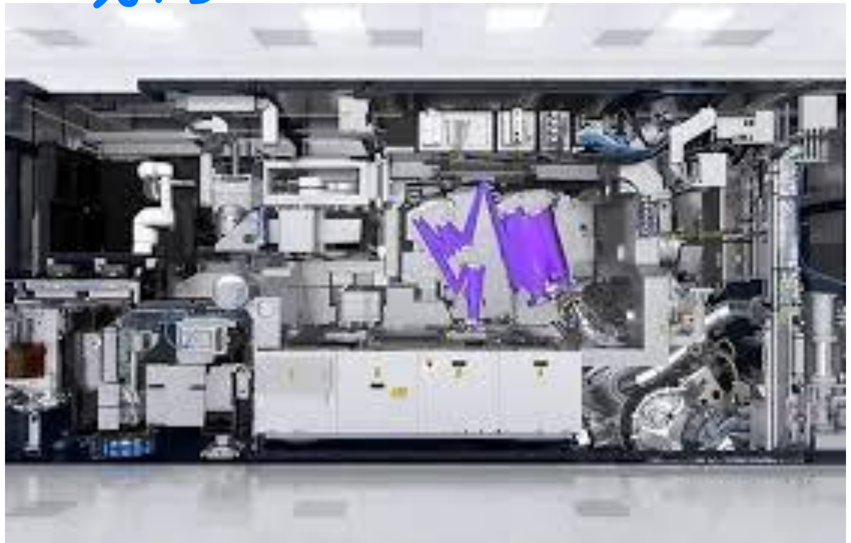
Electromagnetism is useful



电子彈射

mo dem imaging

光刻



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Office Hours

Thursdays, 2:00 to 4:00 pm (appointments needed).

- **Textbooks**

1. Main textbook: *Introduction to Electrodynamics*, David J. Griffiths, Cambridge University Press, Fourth Edition (ISBN: 978-1108420419).
    - (Electronic version downloadable at http://reserves.lib.tsinghua.edu.cn/Home).
2. *The Feynman Lectures on Physics, Volume II*
    - (Electronic version downloadable at https://www.feynmanlectures.caltech.edu/II_toc.html)
3. 《电动力学》郭硕鸿 著，高等教育出版社，第三版(ISBN:978-7040239249).

- **Grading**
- Homework 20%; midterm exam 40%; final exam 40%.
- **A (A, A-): 20%; B (B+, B, B-): 50%; else: 30%.**

Week	Date	Topics
1	Feb 27	Basics of vector analysis.
2	Mar 6	Electrostatics, Coulomb's law, Gauss's law, conductors.
3	Mar 13	Electric potential, dielectric and polarization.
4	Mar 20	Electric boundary conditions, uniqueness theorem, method of images, separation of variables.
5	Mar 27	Magnetostatic fields, steady currents.
6	Apr 3	Properties of magnetostatic fields, vector potential, magnetic fields in matter, magnetic boundary conditions.
7	Apr 10	Force and energy of electric and magnetic dipoles, A-B effect, electromagnetic induction, Faraday's law.
8	Apr 17	Displacement current, Maxwell's equations.
9	Apr 24	Mid-term exam (open-book, 2 hours).
10	May 1	No class
11	May 8	Poynting's theorem, EM waves, plane waves.
12	May 15	General solutions to the wave equations, dipole radiation, antenna.
13	May 22	Linear light-matter interaction, the origin of refractive index, absorption, dispersion.
14	May 29	Reflection and transmission of plane waves at an interface.
15	Jun 5	Waveguides.
16	Jun 12	Course review or no class, depending on the final exam date
17	TBD	Final exam (open-book, 2 hours).

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

Fields

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

Waves

Hints about how to study this course

- Be prepared:
  - Not intuitive (compared to classical mechanics)
  - Very abstract (field, potentials, **vector potential**, waves…)
- **Keys: imagination and mathematics**

Physics

vs.

Math

- **Study tip: Preview the material and bring your questions to class.**

Brief review of basic math：

1. a, b, a+b, ab=ba, a(b+c) = ab + ac, $(ab)c = a(bc)$…

2. $f(x)$, $f'(x)$, $\boxed{\mathrm{d}f(x) = f'(x)dx,}$ $(fg)' = f'g + fg'$…

3. $f''(x) = (f'(x))'$

4. $\int_a^b f'(x)dx = f(b) - f(a)$

$$A = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\vec{A} \cdot \vec{B} = \begin{pmatrix} A_x & A_y & A_z \end{pmatrix} \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix}$$

$$= A_x B_x + A_y B_y + A_z B_z$$

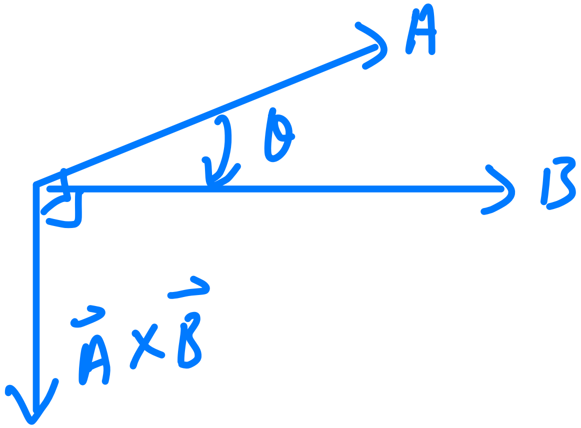
$= \text{scalar}$

$= AB\cos\theta$ if $\theta = 90^\circ$, independent

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$$



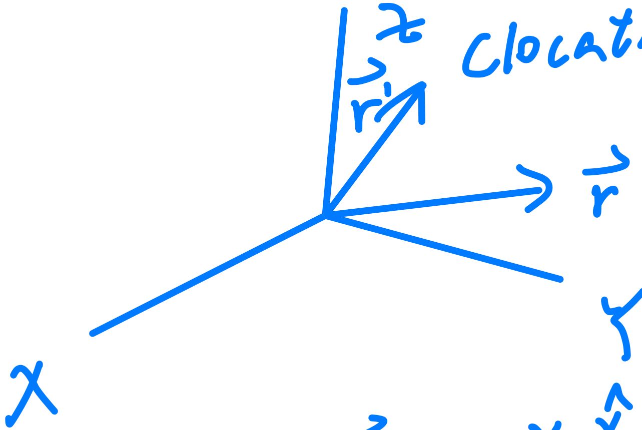
Triple product

$\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$

Rotation!

$\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$

$\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}\cdot(A\cdot\vec{B})$ BAC CAB Rule



(location where you put the source)

(location where you want to evaluate the field!)

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\vec{r}' = x'\hat{x} + y'\hat{y} + z'\hat{z} = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$

$\vec{\mathscr{r}} = \begin{bmatrix} x - x' \\ y - y' \\ z - z' \end{bmatrix}$

$\hat{\mathscr{r}} = \dots$

$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ (divide to components)

Differential Cal.

$f(x) \to f'(x)$

∇ del operator $\triangleq \begin{bmatrix} \partial x \\ \partial y \\ \partial z \end{bmatrix}$

$$\frac{\partial}{\partial x}$$

Gradient

$\nabla f(x,y,z) = \begin{bmatrix} \partial_x f \\ \partial_y f \\ \partial_z f \end{bmatrix}$ vector function

$df(x,y,z) = \partial_x f\,dx + \partial_y f\,dy + \partial_z f\,dz$

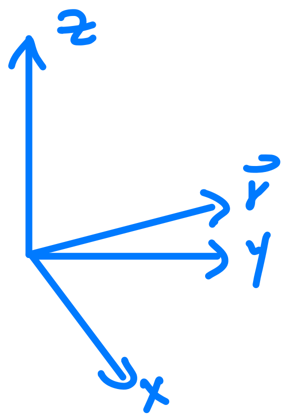
$= (\partial_x f, \partial_y f, \partial_z f) \cdot \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$

$= \nabla f \cdot d\vec{l}$

$\nabla f \cdot d\vec{l}$

along direction, what change is the maximum

$d\vec{l} = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$



1 mark!

$$\nabla |r| = (\partial x, \partial y, \partial z)\left(\sqrt{x^2+y^2+z^2}\right)$$

$$= \left\langle \frac{2x}{2r}, \frac{2y}{2r}, \frac{2z}{2r} \right\rangle$$

$$= \left\langle \frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right\rangle$$

$$= \frac{1}{r}\langle x, y, z \rangle$$

$$= \hat{r}$$

Divergence

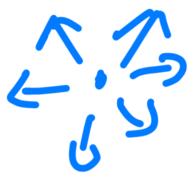
$$\nabla \cdot \vec{f} = (\partial_x, \partial_y, \partial_z) \cdot \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix}$$

$$= \partial_x f_x + \partial_y f_y + \partial_z f_z$$

No term like $\partial_x f_y$

like a fountain

How similar the field like this pattern, so taking dot product.



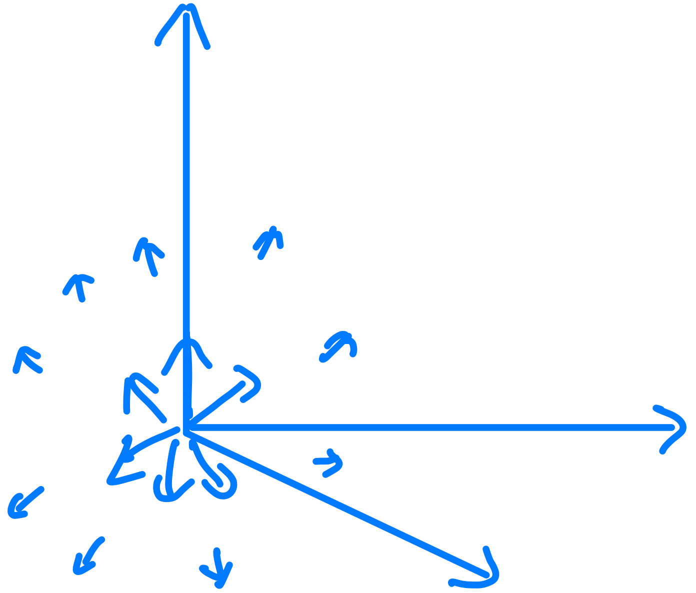
$$\nabla \cdot \frac{\hat{r}}{r^2} =$$

most fundamental function of entire universe!

$$\nabla \frac{\vec{r}}{r^2} = \nabla \cdot \frac{\vec{r}}{r^3}$$

$$= (\partial x, \partial y, \partial z) \cdot \frac{\langle x, y, z \rangle}{r^3}$$

$$= \frac{1}{r^3} - \frac{3}{2}\frac{2x^2}{r^3} + \cdots$$



Curl

$$\nabla \times \vec{f} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial x & \partial y & \partial z \\ f_x & f_y & f_z \end{vmatrix}$$

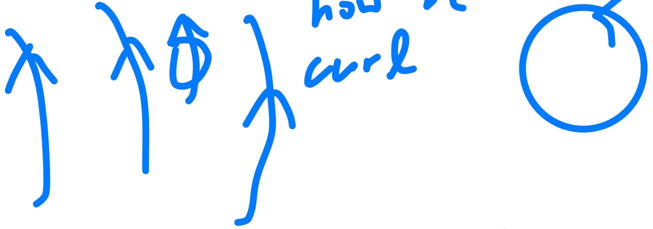
$$= \hat{x}(\partial_y f_x - \partial_z f_y) + \hat{y}(\partial_z f_x - \partial_x f_z) + \hat{z}(\partial_x f_y - \partial_y f_x)$$

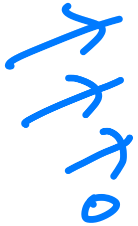
fixed location, how it rotating around?

will it rotate?

how it rotate?

curl

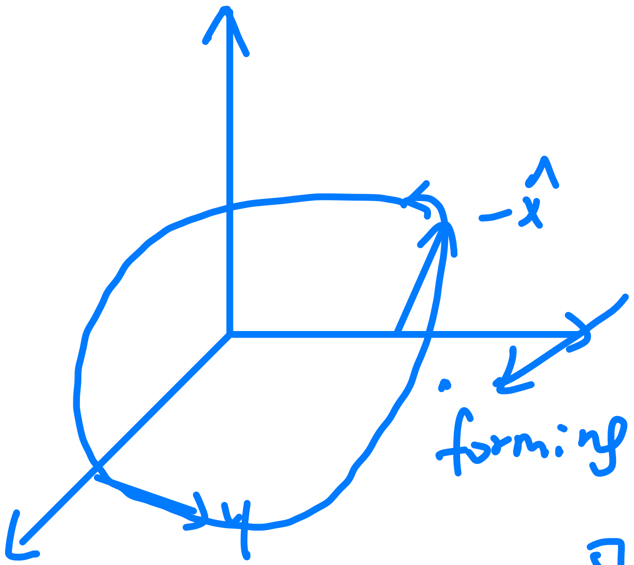




$\nabla \times \vec{f} = 0$

ball will not rotate!

$\vec{f} = x\hat{y} - y\hat{x}$



curl of this location still 2!

forming a loop

$\nabla \times \vec{f} = \cdots \ (\leq 1)$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial x & \partial y & \partial z \\ -y & x & 0 \end{vmatrix} = 2\hat{z}$$

Right hand rule, counterclockwise along x-y plane!

Now move on to 2nd derivative

$f'' = ?$

use del operator

$$\nabla\cdot(\nabla f) = (\partial x \;\; \partial y \;\; \partial z)\begin{bmatrix} \partial x f \\ \partial y f \\ \partial z f \end{bmatrix}$$

$$= \partial xx f + \partial yy f + \partial zz f$$

$$= (\partial xx + \partial yy + \partial zz) f$$

$$= \nabla^2 f$$

scalar operator!

$$\nabla^2 \vec{f} = \begin{bmatrix} \nabla^2 f_x \\ \nabla^2 f_y \\ \nabla^2 f_z \end{bmatrix}$$

$$\nabla \times (\nabla \times \vec{f})$$

$$= \nabla(\nabla \cdot \vec{f}) - \nabla^2 \vec{f}$$ (remember this result)

→ derive wave equation

BAC - CAB rule!

Once a vector function can be represented by gradient by scalar function, curl will be vanished

$$\nabla \times (\nabla f)$$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ f_x & f_y & f_z \end{vmatrix}$$

$$= \hat{x}(\partial_y \partial_z f - \partial_z \partial_y f) + \dots$$

$$= \vec{0}$$

☆ very important!

Because ∇f maybe very complex!

nontrivial

$$\nabla \cdot (\nabla \times \vec{f})$$

$$= (\partial_x \quad \partial_y \quad \partial_z) \begin{bmatrix} \partial_y f_z - \partial_x f_y \\ \partial_z f_x - \partial_x f_z \\ \partial_x f_y - \partial_y f_x \end{bmatrix}$$

$$= \partial_x \partial_y f_z - \partial_x \partial_x f_y + \cdots$$

$$= 0$$

divergence and curl is 0

$\nabla \vec{f}$, ∇ is perpendicular! $\Rightarrow$ no magnetic monopole

Integral Calculus

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

$f'(x)$ may be complex function

- not a trivial result

$$\int_{\vec{a}}^{\vec{b}} \nabla f(x,y,z)\cdot d\vec{l}$$

field, conservative field

$$= \int_{\vec{a}}^{\vec{b}} df = f(\vec{b}) - f(\vec{a})$$

much more interesting

1. Independent on path!
2. Curl of such function $= 0$

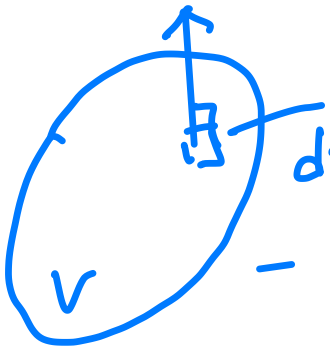
Same final result

$\iiint_V \nabla\cdot\vec{f}\,dV$ volume integration

$\int_V$

$= \oiint_S \vec{f}\cdot d\vec{a}$ = flux

$\oint$ closed surface



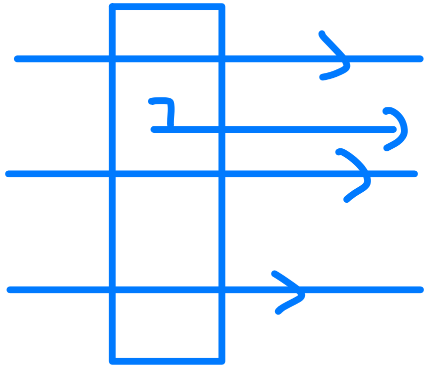
$d\vec{a}$

direction ⊥ local area

V

S, closed surface

But not easy to prove



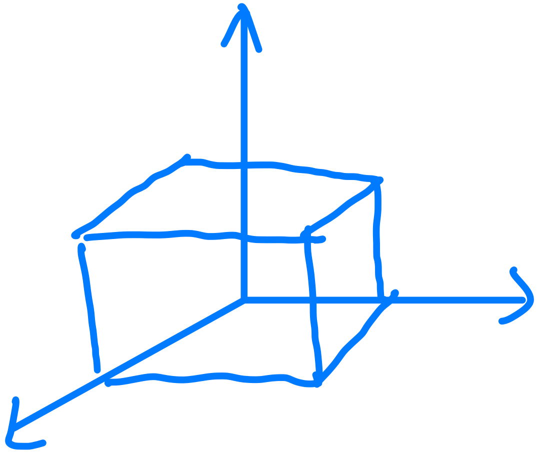
$\int$, Total number of field lines that penetrates the surface

Divergence theorem

$$\int_{z_1}^{z_2}\int_{y_1}^{y_2}\int_{x_1}^{x_2} \nabla \cdot f(x,y,z)\,dx\,dy\,dz = \int_{z_1}^{z_2}\int_{y_1}^{y_2} f(x_2,y,z)\,dy\,dz \; - \int_{z_1}^{z_2}\int_{y_1}^{y_2} f(x_1,y,z)\,dy\,dz + \dots$$

$$\partial_x f_x + \partial_y f_y + \partial_z f_z$$

(4 terms left)



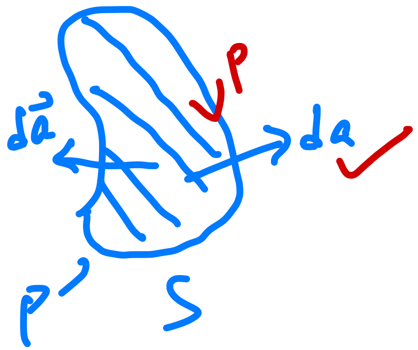
$$= \int_{s_1} \vec{f}\cdot d\vec{a} + \int_{s_2} \vec{f}\cdot d\vec{a} + \dots$$

$dydz$, along $\hat{x}$

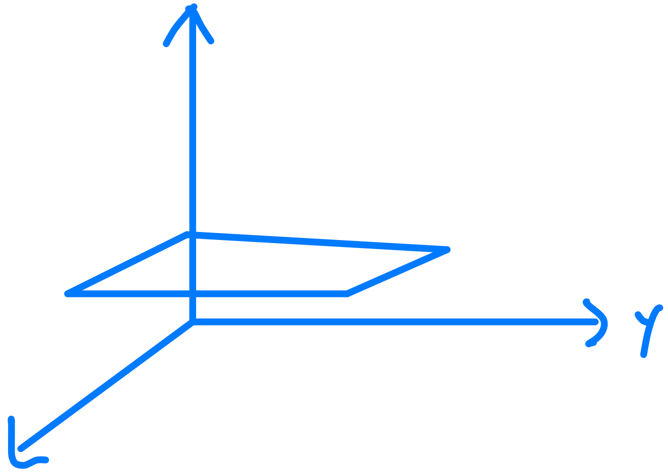
4 surfaces

$$\iint_S (\nabla \times \vec{f}) \cdot d\vec{a} = \oint_P \vec{f} \cdot d\vec{l}$$

S ← open surface



P is defined by right hand rule



$$\nabla \cdot \vec{E} = \iiint_V \nabla \cdot \vec{E}\, dV = \frac{\iiint_V \rho(x,y,z,t)\, dV}{\varepsilon_0} = \frac{Q_{enc}}{\varepsilon_0}$$

Divergence theorem

$$\oiint E \cdot d\vec{a} = \frac{Q_{enc}}{\varepsilon_0}$$

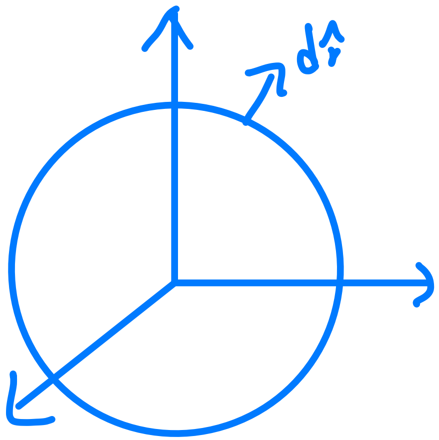
Total flux

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial E}$$

$$\iint_S \nabla \times \vec{E} \cdot d\vec{a} = -\frac{d}{dt}\iint_S \underbrace{B \cdot d\vec{a}}_{\text{flux}} = -\frac{d}{dt}\Phi_B = \underbrace{\oint \vec{E} \cdot d\vec{l}}_{\text{电动势}}$$

- only study local behaviour in flux

$$\iiint_V \nabla \cdot \frac{\hat{r}}{r^2}\, dV = \oiint_S \frac{\hat{r}\cdot\hat{r}}{R^2} \cdot R^2 \sin\theta\, d\theta\, d\phi$$



$$= \int_0^{2\pi}\int_0^{\pi} \sin\theta\, d\theta\, d\phi$$

$$= 2\pi \cdot 2 = 4\pi$$

$\neq 0$

origin is not 0!

$$\left(\nabla \frac{\hat{r}}{r^2}\right) = 4\pi\, \delta^3(\vec{r})$$

origin

$\delta^3(\quad)$

$= \delta^3(\vec{r} - \vec{r}\,')$ shift the source!

impulse

$$\delta(x) = \int \begin{matrix} -\infty & x = 0 \\ 0 & x \neq 0 \end{matrix}$$

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1$$